

# Wonbeen Yoon

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## EDUCATION

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### Seoul National University

Seoul, South Korea

*B.S. in Mechanical Engineering and Electrical and Computer Engineering*

*Mar 2021 – Mar 2028 (Expected)*

*Cumulative GPA: 4.14 / 4.30*

**Relevant Coursework:** Digital Logic Design and Lab, Digital Systems Design and Lab, Embedded System Design, Mechanical System Modeling and Control, Fundamentals of Control Engineering, Introduction to Robotics.

**Research Interests:** Hardware-algorithm co-design for real-time embedded systems, power-efficient RTL/HLS accelerators, FPGA-based on-device processing, hardware-aware robot control and perception, and reliable physical AI systems.

### Gwangju Science Academy

Gwangju, South Korea

*High School Diploma; GPA: 4.10/4.30*

*Mar 2018 – Feb 2021*

## EXPERIENCE

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### Mutual.ai (Startup Project)

Seoul, South Korea

*Hardware Engineering Associate*

*July 2025 – Feb 2026*

- Led the development of an FPGA-based secure camera pipeline for data provenance, focusing on hardware-accelerated image processing and data integrity.
- Designed and verified digital logic circuits in Verilog, managing iterative hardware verification cycles to address timing and integration challenges.
- Collaborated on the end-to-end integration of hardware modules with software application layers to create a secure imaging proof-of-concept.

### Interactive & Networked Robotics Laboratory, Seoul National University

Seoul, South Korea

*Undergraduate Research Intern*

*Dec 2024 – Apr 2025*

- Adapted an existing MATLAB mathematical model to incorporate hybrid-actuation logic for the LASDRA (Large-Size Aerial Skeleton System with Distributed Rotor Actuation) framework.
- Evaluated system performance by sourcing real-world hardware specifications—including motor constants, weights, and dimensions—to define high-fidelity simulation parameters.
- Conducted a comparative power consumption analysis between internal, external, and hybrid-actuated configurations to validate the efficiency of the hybrid architecture.

## SKILLS

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**Programming & Computing:** C/C++, Python, MATLAB, C++ Multithreading, OpenMP, CUDA, ARM NEON SIMD

**Hardware Design:** Verilog, RTL Design, FPGA Implementation, HLS, Hardware/Software Integration

**Languages:** Korean (Native), English (Intermediate)

## AWARDS & ACHIEVEMENTS

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**National Science & Technology Scholarship** (2025-1, 2025-2, 2026-1)

**Academic Scholarship** (2021-2, 2022-1, 2024-2)